20250929EB_MN (vLH103,51)

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2025年9月29日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年10月1日 星期三

T2 -> T3

2025年10月3日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年10月6日 星期一

T4 -> T5

2025年10月8日 星期三

T5 -> T6

2025年10月10日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年10月13日 星期一

T6 -> T7

Dissociation (try 3 methods)

1. Collect EBs in 50mL falcon and split cells into 3 portions.
2. Spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs). (I forgot if I spin it down or just let gravity to that)
3. Add 3 different reagents to dissociate cells. Tap the tube during the incubation. (see the table1)
4. Add 2mL of Heat Inactivated FBS per method into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
5. Add 2mL of CTWM per method and pipette it with 5mL serological pipette gently.
6. Filter the cell with 40um filter into a new 50mL tube
7. Spin the cell down at 200rcf, 3.5min. Remove the supernatent.
8. Add 1mL T7 to resuspend the cells and count it.
9. Plate the cells with T7.

cell info

	A	B	C
1	cell	passage	well#
2	WT11	P6	5

Table1

	A	B	C	D	E	F	G
1		reagent	volume	treat time	temp	cell count (K/mL, 1mL in total)	survival
2	method 1	accutase	1mL	50min	RT	607	92
3	method 2	0.05% trypsin	2mL	16min	37C	398	100
4	method 3	0.025% trypsin	2mL	26min	37C	1070	100

Table3

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	691.6666666667	NA	3	2075	A	24	259.375	8	Add 1mL more and distribute to 8 wells evenly

Plate A

	1	2	3	4	5	6
A	vLH51, 2uL in total (0.5uL/well)	vLH103, 3uL in total (0.75uL/well)				
B						
C						
D						

2025年10月14日 星期二

Change T7 -> T7 if added virus

2025年10月16日 星期四

T7 -> T8

2025年10月17日 星期五

1/2 T8 -> T8

2025年10月20日 星期一

1/2 T8 -> 1/2 T8

2025年10月22日 星期三

~5pm Collect 200uL from B1 and B2 before changing media --> glucose concentration assay.
1/2 T8 -> 1/2 T8

2025年10月23日 星期四

treat 24h timepoint at 11:00 am (C2D2) and 12:00pm (C1D1)
change half media for the rest --> collect media before changing media **--> glucose concentration assay.**

2025年10月24日 星期五

microscope: new spinning disk confocal
100X, 1xzoom, 0.11um/px
The new scope switch back to one camera system on Monday. This is the first time using this new setting.

Table2

	A	B	C	D	E	F
1			103		51	
2				image		image
3	1	2.5mM 1h	10:00	11:00	8:45	cells look not healthy --> didn't take the images
4	2	0.15mM 1h	10:30	11:30	9:15	Since I didn't take control cells, I skipped this one.
5	3	2.5mM 24h	10/23 11:00	11:55	10/23 12:00	12:30
6	4	0.15mM 24h	10/23 11:00	12:11	10/23 12:00	

vLH103

	A	B	C	D
1		laser power	exposure time	
2	488 10ms	100	10ms	movie
3	640	10	100ms	snapshot
4	488 search	30	50ms	snapshot

vLH103_fileid-conditions

	A	B	C	D
1		treatment	treat time	type
2	vLH103_MN_100x_1xzoom_0001	2.5mM gluc	1h	movie
3	vLH103_MN_100x_1xzoom_0002	2.5mM gluc	1h	snapshot
4	vLH103_MN_100x_1xzoom_0003	0.15mM gluc	1h	movie
5	vLH103_MN_100x_1xzoom_0004	0.15mM gluc	1h	snapshot
6	vLH103_MN_100x_1xzoom_0005	2.5mM gluc	24h	movie
7	vLH103_MN_100x_1xzoom_0006	2.5mM gluc	24h	snapshot
8	vLH103_MN_100x_1xzoom_0007	0.15mM gluc	24h	movie
9	vLH103_MN_100x_1xzoom_0008	0.15mM gluc	24h	snapshot

I use JOBS for vLH103. Pick multiple points first and take the movies and snapshot all together.
I did the same setting for the soma and neurites of C1. However, I found the neurite bleach easily. Therefore, I switch to ND acquisition to take the movie whenever I found one for C2. The OC is the same, but C1 and C2 is different.

vLH51

	A	B	C	D
1		laser power	exposure time	
2	488 10ms	100	10ms	movie

vLH51_fileid-conditions

	A	B	C	D
1		treatment	treat time	type
2	soma_vLH51_MN_100x_1xzoom_XY01	2.5mM gluc	24h	movie
3	soma_vLH51_MN_100x_1xzoom_XY02	2.5mM gluc	24h	movie
4	soma_vLH51_MN_100x_1xzoom_XY03	2.5mM gluc	24h	movie
5	soma_vLH51_MN_100x_1xzoom_XY04	2.5mM gluc	24h	movie
6	soma_vLH51_MN_100x_1xzoom_XY05	2.5mM gluc	24h	movie
7	soma_vLH51_MN_100x_1xzoom_XY06	2.5mM gluc	24h	movie
8	soma_vLH51_MN_100x_1xzoom_XY07	2.5mM gluc	24h	movie
9	soma_vLH51_MN_100x_1xzoom_XY08	2.5mM gluc	24h	movie
10	soma_vLH51_MN_100x_1xzoom_XY09	2.5mM gluc	24h	movie
11	soma_vLH51_MN_100x_1xzoom_XY10	2.5mM gluc	24h	movie
12	soma_vLH51_MN_100x_1xzoom_XY11	2.5mM gluc	24h	movie
13	soma_vLH51_MN_100x_1xzoom_XY12	2.5mM gluc	24h	movie
14	soma_vLH51_MN_100x_1xzoom_XY13	2.5mM gluc	24h	movie
15	soma_vLH51_MN_100x_1xzoom_XY14	2.5mM gluc	24h	movie
16	soma_vLH51_MN_100x_1xzoom_XY15	2.5mM gluc	24h	movie
17	soma_vLH51_MN_100x_1xzoom_016	0.15mM gluc	24h	movie
18	soma_vLH51_MN_100x_1xzoom_017	0.15mM gluc	24h	movie
19	soma_vLH51_MN_100x_1xzoom_018	0.15mM gluc	24h	movie
20	soma_vLH51_MN_100x_1xzoom_019	0.15mM gluc	24h	movie
21	soma_vLH51_MN_100x_1xzoom_020	0.15mM gluc	24h	movie
22	soma_vLH51_MN_100x_1xzoom_021	0.15mM gluc	24h	movie
23	soma_vLH51_MN_100x_1xzoom_022	0.15mM gluc	24h	movie
24	soma_vLH51_MN_100x_1xzoom_023	0.15mM gluc	24h	movie
25	soma_vLH51_MN_100x_1xzoom_024	0.15mM gluc	24h	movie
26	soma_vLH51_MN_100x_1xzoom_025	0.15mM gluc	24h	movie
27	soma_vLH51_MN_100x_1xzoom_026	0.15mM gluc	24h	movie
28	soma_vLH51_MN_100x_1xzoom_027	0.15mM gluc	24h	movie
29	soma_vLH51_MN_100x_1xzoom_028	0.15mM gluc	24h	movie
30	soma_vLH51_MN_100x_1xzoom_029	0.15mM gluc	24h	movie
31	soma_vLH51_MN_100x_1xzoom_030	0.15mM gluc	24h	movie
32	neurite_vLH51_MN_100x_1xzoom_XY01	2.5mM gluc	24h	movie
33	neurite_vLH51_MN_100x_1xzoom_XY02	2.5mM gluc	24h	movie

34	neurite_vLH51_MN_100x_1xzoom_XY03	2.5mM gluc	24h	movie
35	neurite_vLH51_MN_100x_1xzoom_XY04	2.5mM gluc	24h	movie
36	neurite_vLH51_MN_100x_1xzoom_XY05	2.5mM gluc	24h	movie
37	neurite_vLH51_MN_100x_1xzoom_XY06	2.5mM gluc	24h	movie
38	neurite_vLH51_MN_100x_1xzoom_XY07	2.5mM gluc	24h	movie
39	neurite_vLH51_MN_100x_1xzoom_XY08	2.5mM gluc	24h	movie
40	neurite_vLH51_MN_100x_1xzoom_XY09	2.5mM gluc	24h	movie
41	neurite_vLH51_MN_100x_1xzoom_XY10	2.5mM gluc	24h	movie
42	neurite_vLH51_MN_100x_1xzoom_011	0.15mM gluc	24h	movie
43	neurite_vLH51_MN_100x_1xzoom_012	0.15mM gluc	24h	movie
44	neurite_vLH51_MN_100x_1xzoom_013	0.15mM gluc	24h	movie
45	neurite_vLH51_MN_100x_1xzoom_014	0.15mM gluc	24h	movie
46	neurite_vLH51_MN_100x_1xzoom_015	0.15mM gluc	24h	movie
47	neurite_vLH51_MN_100x_1xzoom_016	0.15mM gluc	24h	movie
48	neurite_vLH51_MN_100x_1xzoom_017	0.15mM gluc	24h	movie